# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The twinkling of stars is due to 1

**(a)** atmospheric refraction of starlight **(b)** reflection by the moon
**(c)** scattering **(d)** dispersion

**2.** When the eye looks at a nearby object, the ciliary muscles 1

**(a)** contract and the lens becomes thinner **(b)** relax and the lens becomes thinner
**(c)** do not act **(d)** contract and the lens becomes thicker

**3.** A person with a far point at 2 m needs a 1

**(a)** convex lens of power +0.5 D **(b)** concave lens of power -0.5 D
**(c)** convex lens of power +2 D **(d)** concave lens of power -2 D

**4.** Planets do not twinkle because 1

**(a)** they are far away
**(b)** they are small
**(c)** they are nearer and act as extended sources
**(d)** they have their own light

**5.** Which part of the eye converts light into electrical signals? 1

**(a)** Cornea **(b)** Iris **(c)** Lens **(d)** Retina

**6.** Cause of hypermetropia may be 1

**(a)** eyeball too long **(b)** cornea too curved
**(c)** lens too thick **(d)** eyeball too short

**7.** The Sun appears slightly higher than its actual position at sunrise because of 1

**(a)** atmospheric refraction **(b)** reflection
**(c)** scattering **(d)** dispersion

**8.** Persons suffering from myopia cannot see 1

**(a)** nearby objects clearly **(b)** distant objects clearly
**(c)** coloured objects only **(d)** any object

**9.** The lens power in dioptres for a lens of focal length -50 cm is 1

**(a)** +2 D **(b)** +0.5 D **(c)** -2 D **(d)** -0.5 D

**10.** Advance sunrise and delayed sunset is about 1

**(a)** 2 hours **(b)** 20 minutes **(c)** 20 seconds **(d)** 2 minutes

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Why do we see a rainbow only after a rain shower and with the Sun behind us? 2

**12.** Why does the eye lens need to change its focal length? 2

**13.** Why is the sky blue? 2

**14.** Why does the Sun look red at sunrise? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A person cannot see objects beyond 1.2 m. What defect does he have and what lens power will correct it? 3

**16.** Explain how a rainbow is formed. 3

**17.** What do you mean by the Tyndall effect? Give an example. 3

OR

Why can a person with myopia see nearby objects clearly? 3

**18.** What is atmospheric refraction? Give two examples of its effects. 3

OR

Why do danger signals use red light? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A person has a near point at 1 m. Find the power of the lens required to read a book at 25 cm. 5

OR

Explain why 5
(a) the sky looks dark to an astronaut
(b) danger signals are red
(c) the Sun appears flattened at sunset.

**20.** A myopic person has a far point at 80 cm. Find the focal length and the power of the corrective lens. 5

OR

What is the Tyndall effect? Describe the factors on which the colour of scattered light depends. 5

## END OF QUESTION PAPER — Total: 40 Marks

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